

QUANTITATIVE RESEARCH

Ghost Whales in Biotech Options

Option flow, public news and the cost of protection

Research conclusion. The registered strategy fails after costs. Large directional option prints precede positive average signed stock returns, but the estimates are uncertain and concentrated in a few trades. The options hedge consumes the apparent edge. This is evidence against deploying the tested strategy.

1 Economic hypothesis and contribution

Biotech prices respond sharply to trial readouts, regulatory decisions and acquisitions. We hypothesize that large directional option trades can reveal information before it reaches stock prices. The potential counterparties are option market makers and investors who have not yet incorporated that information. Market-maker hedging may reinforce a directional move, although print direction alone cannot establish informed trading [1-5].

The proposed persistence mechanism is costly interpretation of fragmented option flow and uneven public-news coverage. Competing flow traders and faster dissemination could remove the opportunity. Options provide leverage; their use is not evidence that a trader possesses private information.

Contribution. Sign public OPRA prints against contemporaneous quotes, condition the signal on observed news sentiment, and measure the economics of an options tail hedge. The comparison tests whether flow contains incremental predictive information beyond the measured news filter; it does not directly identify private information.

Registered predictions. P1: uncovered whale days have positive net five-session returns. P2: news-covered days have little predictive return. P3: the edge increases with whale-premium size. P4: it fades within 10-20 sessions. Failure conditions include negative net OOS performance, unstable nearby parameters, factor-explained returns or failure at doubled costs.

2 Data and research record

Massive supplies the volume screen, sentiment, option bars and liquidity inputs; Databento supplies OPRA trades with quotes; Webull supplies stock and XBI execution bars. The universe contains 1,061 mapped biotech symbols, including dead/acquired names; 599 had listed options. OPRA was purchased for 2,452 sampled name-days at a reported cost of \$63.52.

The source note records registration at commit **16e75ab** (Oct 3, 18:39 ET), IS results at **3c468f6**, and an OOS run at 22:12 ET. There were 23 reported configurations: 16 IS and seven predeclared OOS evaluations. Registration and reproduction remain subject to code review.

3 Signal, portfolio and test design

Screen. Flag sampled option volume at least twice its 20-session median. Apply contract-count and directional-share screens, stock price of at least \$3, and prior 20-day dollar volume of at least \$5M. The source reports 71,294 initial name-days and 7,976 after screening.

Sign and select. A seeded 30% sample (seed 20261003) provides 2,452 days for OPRA analysis. Sign prints of at least \$50,000: at/above ask is bought, at/below bid is sold, and interior prints use the nearer quote. Bullish premium is calls bought plus puts sold. Require a 3x volume spike, at least 1,000 full-tape contracts, and 65% of whale premium on one side: 953 days.

News and entry. Skip days with same-direction ticker sentiment from three days before the signal through the following open: 857 signals remain. Enter stock at the next open and exit at the open five sessions later. Of these signals, 650 completed trades, 203 were skipped and four arrived too late to complete the hold. Skips comprised 102 missing bars, 63 already-held names and 38 position-cap exclusions.

Portfolio rules. Weight = $\min(1\% / \text{trailing 20-day daily volatility}, 10\%)$; cap orders at 2% of trailing share ADV and hold at most 10 names. Hedge net stock dollars daily with XBI. Exit a 15% close-to-entry loss at the next open. Halve new sizes below a 15% equity drawdown and restore on recovery. Dollar hedging does not guarantee beta neutrality.

Options hedge. Buy a roughly 10% OTM put on longs or call on shorts, approximately 30 days to expiry; sell when the stock exits. The implementation uses entry-day option VWAP plus 5% of premium. This whole-day observation is unavailable at the stock open: it is an execution approximation, not a demonstrated opening fill. Hedge availability and intraday protection therefore need timestamp-level validation.

Costs. Stock execution costs 20 bps per side; XBI costs 2 bps. Annual borrow is 5% on stocks and 0.5% on XBI. The reported doubled-cost case uses 40 bps per stock side. A noisy daily-bar estimate suggested a 33 bps half-spread IS; baseline costs may be optimistic. Exact stress multipliers should be checked against configuration.

Split and limitations. History starts Jan 2, 2024 and ends Oct 2, 2026. The designated OOS window is Mar 16-Oct 2, 2026 (139 sessions), approximately the last 20%. IS signals stop Mar 13, but the final IS position exits Mar 23: IS outcomes overlap the OOS calendar window. This prevents claiming fully disjoint return samples without qualification.

Data handling. Vendor bars are split-adjusted; gaps are not filled. Mid-hold disappearance is valued at the last close, which may not be executable. Missing Webull names generated 86 IS/OOS signals; a Massive-based diagnostic found negative average signed returns for these names. Exclusion can flatter reported performance.

4 Results: the registered strategy fails

Metric	IS baseline	IS 2x	OOS baseline	OOS 2x
Ann. return	-11.5%	-18.6%	-25.7%	-31.3%
Volatility	18.8%	17.6%	32.3%	30.2%
Sharpe	-0.56	-1.08	-0.76	-1.09
Max drawdown	-27.8%	-38.9%	-27.3%	-30.1%
Turnover / yr	46x	44x	57x	54x
Worst month	-7.7%	-7.2%	-10.0%	-12.2%
Trades	501	501	149	149
Hit rate	50.7%	50.7%	51.7%	51.7%

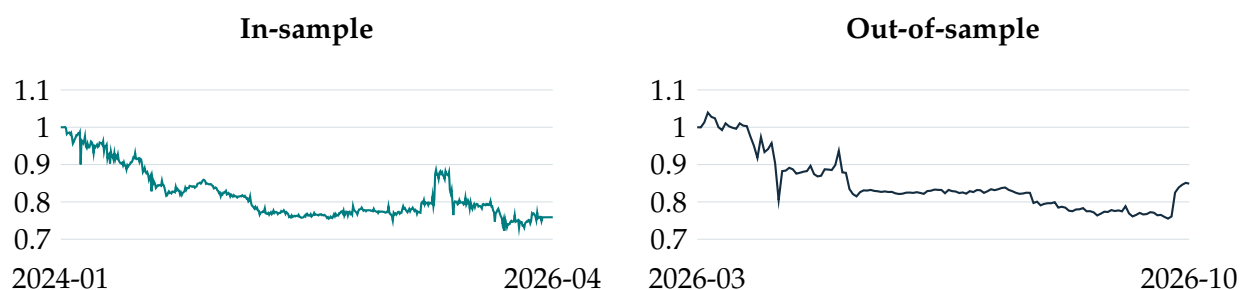


Figure 1. Baseline daily equity, rebased to 1 separately in each sample, from the saved equity CSVs. OOS total return is -15.1%; its annualized return is -25.7%. Turnover includes stock, XBI and options notional divided by mean equity.

P1 fails net. Gross signed five-session signal returns average +1.06% IS (95% CI -0.01% to +2.32%, n=593) and +2.54% OOS (-0.38% to +5.98%, n=173). Both intervals include zero. These event averages differ from sized portfolio returns.

P2-P4 do not validate the thesis. News-covered returns reverse from +3.90% IS to -3.83% OOS. Size-tercile returns rise monotonically only OOS. Returns do not fade by 10-20 sessions; OOS averages are +2.40% at 10 and +3.75% at 20 sessions.

Robustness. All nearby IS parameter variants lose except the \$25k print threshold (Sharpe +0.04). Reported IS XBI beta is 0.03; OOS SPY beta is -0.47. Reported Deflated Sharpe probability is 0.07 in each period. Inference requires review of trial correlation and dependence from overlapping events; these statistics do not establish skill.

5 Attribution and risk management

Hedge economics. Median initial option premium was 4.4% of stock position value (90th percentile: 14%). Net realized hedge cost averaged 0.85% per hedged trade IS and 2.27% OOS. Size-weighted costs of 0.54% and 1.70% exceed the corresponding gross edge of 0.40% and 1.44%, before round-trip stock costs.

Annual return	IS	OOS
No hedges	-3.6%	+50.8%
Add XBI hedge	-4.7%	+18.8%
Add options hedge	-11.5%	-25.7%

The hedge-free variants were declared before OOS, according to the source note. Choosing one after seeing these outcomes would require a new test. The attribution above is sensitive to the hedge execution approximation described on page 2.

Concentration. Long trades average +1.39% gross IS and +5.28% OOS; short trades average +0.01% and -0.03%. These are sample averages, not proof that only bullish prints carry information. The five best trades contribute 142% and 106% of gross P&L. Removing them makes the mean negative. Two OOS longs, KOD and AVTX, contribute 101% of size-weighted gross P&L.

Exposure and drawdown rules. Entry allocations permit up to 100% gross single-stock exposure; XBI and option legs add exposure beyond that amount. Median stock exposure while invested is 25% IS and 30% OOS. Stock net exposure ranges from -80% to +64% before the XBI offset. Drawdown de-risking is active in 77% of sessions in each sample; the stop fires 18 times IS and nine OOS.

Observed tail protection. FDMT loses 67.6% gross versus 20.2% with the modeled hedge, approximately 6.8% versus 2.0% of equity. A close-based stop cannot prevent an overnight gap. A protective option can limit loss only when acquired, retained and executable under the assumed terms; the observed result is not a guaranteed portfolio loss bound.

Coverage failure. No chosen option traded on 32% of IS entries and 27% of OOS entries, leaving those trades unhedged. The worst OOS trade, BIOA (-54.7%, or -2.7% of equity), was among them. Losses also depend on regime: 2024 returns were -23.3%, while 2025 was approximately flat.

Proposed controls, untested. A future specification could limit unhedged names to 5%, add a 20% drawdown kill switch, and test long-only exposure with catalyst-conditional hedging. These are research proposals informed by observed results, not improvements demonstrated on an untouched sample.

6 Liquidity and deployable capital

Baseline stock participation: median 0.10% of ADV, 95th percentile 0.5-0.7%, maximum 2%. Minimum dollar ADV: \$5M. Assumed impact per side = daily volatility x $\sqrt{\text{shares} / \text{ADV}}$, with coefficient 1; this is not calibrated execution evidence [6].

AUM	Participation*	Impact, round trip	Net trade IS / OOS
\$1M	0.1%	29-32 bps	-0.02% / +1.96%
\$5M	0.5%	64-72 bps	-0.38% / +1.56%
\$25M	2.2-2.5%	138-146 bps	-1.12% / +0.82%
\$100M	8.9-9.9%	211-222 bps	-1.84% / +0.06%

*Scaling scenario: participation above 2% is not evidence of compliant fills; pre-/post-cap aggregation needs clarification. Net trade deducts 40 bps round-trip cost plus impact, excluding the tail hedge.

Capacity conclusion. There is no demonstrated profitable capacity for the tested strategy. Even the hedge-free IS scenario is negative at \$1M after assumed impact. The favorable OOS scenario depends heavily on two outliers. Option liquidity, unavailable contracts and potentially expensive or unavailable biotech borrow impose additional constraints; a profitable long-only capacity estimate remains untested.

7 Limitations and next research decision

Hedge expense, weak short-side averages, unstable news filtering and concentration describe the failure. The 30% sample and 139-session OOS window leave substantial uncertainty. Quote signing can misclassify multi-leg activity; missing sentiment does not establish absence of news.

Resolve hedge VWAP and availability, IS/OOS overlap, delisting valuation and reproducibility. Inference should account for date and issuer dependence. All 23 configurations are disclosed; the attractive hedge-free OOS outcome is not adopted.

Next test. Register a long-only, catalyst-conditional-hedge, 10-session specification and evaluate future data without retuning. The unsigned 70% of screened historical days offers additional contemporaneous validation, but shares regimes and possibly issuers with the inspected sample; it is not a fresh temporal holdout.

8 Reproduction and conclusion

From tracks/whale_footprints: install requirements.txt; configure .env; run `python run_all.py --from-signals` (add `--oos` for OOS), then `python note_stats.py` and `python make_figures.py`. Vendor credentials may be needed. This editorial revision has not rerun the pipeline.

Decision: do not deploy the tested strategy. Preserve the negative result and treat the long-side pattern as a hypothesis requiring new evidence.

References

Outside the five-page main note.

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Data and software. Databento OPRA; Massive option, stock and news data; Webull stock/ETF bars. Research software: backtrader, pandas, numpy, scipy and matplotlib. Bibliographic details and reported results are carried from the supplied research note.

Document provenance. Condensed and reorganized from QUANT_NOTE.md. No new backtest was run and no result was selected to improve performance. Figure 1 was redrawn from saved `is_equity.csv` and `oos_equity.csv`. This edition makes execution and sample-boundary qualifications explicit.